THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS TWO-CHAPTER TEST SET D

Chapters 9 & 10: **Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these has rotational symmetry of order 2 but not 4? 1

(a) Square **(b)** Equilateral triangle
(c) Rectangle **(d)** Circle

2. A kite has: 1

(a) 2 lines **(b)** 4 lines
(c) no line **(d)** 1 line of symmetry

3. When a figure is reflected in a mirror, its size: 1

(a) halves **(b)** stays the same **(c)** changes randomly **(d)** doubles

4. The order of rotational symmetry of a regular pentagon is: 1

(a) 10 **(b)** 1 **(c)** 2 **(d)** 5

5. The number of lines of symmetry of a regular heptagon is: 1

(a) 1 **(b)** 7 **(c)** 14 **(d)** 3

6. A lift moves from the ground floor to 3 floors below. The integer is: 1

(a) 3 **(b)** −3 **(c)** 0 **(d)** −1

7. If the temperature falls from 2°C by 6°C it becomes: 1

(a) 8°C **(b)** −4°C **(c)** 4°C **(d)** −8°C

8. The value of $-6 + (-4)$ is: 1

(a) 10 **(b)** −12 **(c)** −10 **(d)** −8

9. The value of $12 + (-15)$ is: 1

(a) −3 **(b)** −1 **(c)** 3 **(d)** −5

10. The value of $4 + (-10)$ is: 1

(a) −8 **(b)** −6 **(c)** 6 **(d)** −4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State one use of symmetry in architecture. 2

12. Does a scalene triangle have rotational symmetry? 2

13. A diver is at −12 m and goes 5 m up. Where is she now? 2

14. Find the integer lying midway between −6 and 2. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. The numeral 8 and 0 are written. Which have line symmetry? 3

16. The height of a kite is 40 m above and a fish is 15 m below the ground (water). Find the vertical gap. 3

17. Draw the shape of a pentagon with all its lines of symmetry. 3

OR

Which of the words MOM, BOB, WOW, TOT read the same in a mirror placed vertically beside them? Give reasons. 3

18. Ravi walks 8 steps east (+) then 11 steps west (−). Where is he from the start? 3

OR

Using tokens explain $(-4) - (-6)$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) How are symmetry and reflection related? 5
(ii) Give two examples of rotational symmetry in daily life.

OR

(i) Explain the symmetry of a regular polygon with n sides. 5
(ii) Apply for n = 8.

20. (i) Mention one historical fact about integers in India. 5
(ii) Why were negative numbers needed?

OR

(i) Draw a number line and show 3 + (−5). 5
(ii) Verify by tokens.
(iii) Write a story for it.

END OF QUESTION PAPER Total: 40 Marks

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